MIL-L-20703D
9 May 1985
SUPERSEDING
MIL-L-20703C
28 September 1973

MILITARY SPECIFICATION

LIGHT, IGNITION TIMING (MAGNETO)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of timing light (magneto) that visually indicates, either singly or simultaneously, the timing of magnetos in single or dual ignition systems.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

W-C-440	-	Clips, Electrical, General Specification For.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-C-843	-	Cushioning Material, Cellulosic.
PPP-T-60	-	Tape, Packaging, Waterproof.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-ETS (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4910

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MIL-P-116	-	Preservation, Methods of.
MIL-T-152	-	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-V-173	-	Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-L-15098	-	Lamp, Glow, General Specification for.
MIL-L-15098/5	-	Lamp, Glow, Indicator, Types J2A and B7A.
MIL-L-15098/10	-	Lamp, Glow, Indicator, Types B1A, B1AR, B2A, and B2AR.

STANDARDS

FEDERAL

FED-STD-H28	-	Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods.

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

US DEPARTMENT OF LABOR

OSHA 2206	-	General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish one or more timing lights for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

* 3.2 Design. The timing light shall be new and of the manufacturer's current design meeting the performance requirements specified herein. The light shall be designed to operate on a power source consisting of three, or less, size D dry cell batteries. The case shall be designed to hold the batteries securely in place. (Refer to Figure 1 for general configuration.) Those parts of the light subject to wear, breakage, or distortion shall be accessible for adjustment, repair, and replacement without modification, distortion, or damage to the light or any component.

* 3.2.1 Safety and health requirements. The timing light shall comply with the general safety and health requirements promulgated under OSHA 2206 that are applicable to the light itself. The timing light shall, in addition, comply with the applicable requirements of MIL-STD-454. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 6.2.1 and 6.4).

* 3.2.2 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

* 3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

* 3.3.1 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same

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chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

* 3.4 Construction. The timing light shall be furnished complete so that it can be used for the operations specified herein. Parts used in fabricating the light shall be new and free of defects. All part surfaces shall be clean and free of extraneous materials. External surfaces shall be smooth and free of sharp edges. Functional surfaces shall be free from burrs and defects affecting performance.

* 3.4.1 Electrical connections. Connections to screw-type terminals shall be mechanically secured with means to prevent loss of tightness. All other electrical connections shall be mechanically secured and soldered or brazed. Only rosin base fluxes shall be used in all soldering operations.

* 3.4.2 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness and adjustment. The methods for preventing loss of tightness or adjustment shall be in accordance with accepted engineering standards and practice. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

* 3.4.3 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

3.5 Performance and product characteristics.

3.5.1 Performance. The timing light shall be capable of indicating the timing of one or two magnetos, singly or in pairs, and shall indicate the synchronization of two magneto breakers when used in a dual ignition system.

3.5.2 Endurance.

3.5.2.1 Continuous operation. The timing light shall be capable of continuous operation, with one set of batteries (see 3.2 and 3.7), without a malfunction for a minimum of 2.5 hours.

3.5.2.2 Intermittent operation. The timing light shall be capable of intermittent operation (one minute on and 15 seconds off), with one set of batteries (see 3.2 and 3.7), without a malfunction for a minimum of eight hours.

3.5.3 Vibration. No malfunction or damage to the timing light, or components, shall result from being subjected to a vibration of simple harmonic motion at a frequency of 5 to 55 hertz (Hz), with a maximum amplitude of 0.030 inch (0.060 inch maximum total excursion), for a period of one hour in the direction of each of its three major axes.

3.5.4 Shock. The timing light, with cover in place, shall withstand without damage the shock resulting from being dropped, once through each of its three major axes, a vertical distance of eight inches to a concrete surface. Parts shall not become loosened or damaged.

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3.5.5 Operating temperatures. The timing light shall withstand without damage and shall meet the requirements of 3.5.1 while being subjected to ambient temperatures between +30° ±5° Fahrenheit (F) and +110 ±5°F.

3.6 Components. The timing light shall consist of the components in 3.6.1 through 3.6.2.4 and such other components necessary to provide a device that shall meet the performance requirements specified herein.

3.6.1 Case. The timing light case shall be manufactured of sheet metal that is not less than Manufacturer's Standard Gage No. 20 (0.036 inch). The dimensions, not including handles, hinges, catches, rubber bumpers, and other projections, shall not exceed the following:

(a) Height - 8.5 inches (outside).
(b) Width - 4.5 inches (outside).
(c) Depth - 3.5 inches (outside, including cover).

A label showing schematic wiring shall be affixed to an inside panel surface of the case in a position where the label will be free from damage. A handle shall be provided at the top of the case (see Figure 1). Rubber rest buttons shall be provided at each corner of the back of the case and at each corner of the side (bottom) opposite the carrying handle.

3.6.1.1 Face plate. The general external appearance of the face plate shall be as shown in Figure 1. The face plate shall be attached to the case by at least four mounting screws or other equally effective means.

3.6.1.2 Lead compartment. A compartment for storage of leads (see 3.6.2.3), shall be provided above the top of the face plate as shown in Figure 1.

3.6.1.3 Cover. The cover shall be made of the same material as the case (see 3.6.1) and shall be attached to the case by hinges of a type allowing removal of the cover when the timing light is in use. A snap catch shall be provided that is capable of keeping the cover closed while the timing light is being transported by the handle. The cover, when in place, shall completely cover and enclose the face plate and lead compartment side. A clip shall be provided on the inside of the cover to hold two copies of a handbook of instructions. A label giving operating instructions shall be affixed to the inside surface of the cover.

3.6.2 Circuit parts. All circuit parts shall be assembled or attached to the face plate, or to a chassis attached to the face plate, in such a manner that the entire assembly can be removed from the case by removing the face plate.

3.6.2.1 Circuitry. The circuit shall be designed and constructed to operate with a battery power supply and be operative within a range of 3.0 to 4.5 volts. Maximum current at 4.5 volts shall not exceed 0.150 amperes. No system incorporating indicating lights in the battery circuit shall be acceptable. The circuit constants shall be of such values that, in a temperature of +70° to +90°F, the circuit shall not fire a three point spark gap set at 0.010 inch, with a magneto operating at a speed equivalent to a 4-cycle engine speed of 1,000 revolutions per minute (rpm).

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3.6.2.2 Switch. An on-off switch with visible position markings shall be provided.

3.6.2.3 Test leads. Test leads shall be anchored to the face plate in a manner that shall protect against strain at electrical connections. Leads passing through metal enclosures shall be protected by grommets. When specified (see 6.2.1), an optional method shall be to terminate the test leads with pins fitting into properly marked pin jacks on the face plate (see Figure 1). All test leads shall be single conductor cable and shall be properly identified. Leads shall be a minimum of 48 inches in length. The free ends shall be terminated with insulated clips conforming to type CC of W-C-440. One ground lead and two magneto leads shall be furnished as follows:

(a) The ground lead shall be a single conductor cable, AWG 18 gage, 65 strands, 300 volt rating, oil resistant, neoprene covered type. The lead clip shall be equipped with a black sleeve insulator.
(b) The magneto leads shall be the same as (a) above except that the sleeve insulators shall be red instead of black.

* 3.6.2.4 Glow lamps. The glow lamps shall conform to MIL-L-15098, Military Part Numbers M15098/5-002 or M15098/10-002. The lamps shall be mounted back of the face plate, to the rear of the windows. The lamps shall be fitted with grommets in a manner to project through the windows and to allow for ease of replacement. Circuitry shall be so arranged that both electrodes shall glow. The glowing electrodes shall be visible in daylight.

3.7 Batteries. The timing light shall be designed to use three, or less, size D dry cell batteries. The chassis shall be constructed so that no tools shall be required for installing or replacing batteries. Batteries shall not be supplied with the timing light.

* 3.8 Fungus control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

* 3.9 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

* 3.10 Product identification. The timing light shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number. A nameplate shall be attached securely to the face plate as shown in Figure 1.

* 3.11 Workmanship. Standards of workmanship shall assure that the timing light shall have the stability, strength, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (c) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Functional test (see 4.6.1).
(c) Packaging inspection (see 4.7).

* 4.4.1 Sampling. Unless otherwise specified (see 6.2.1), sampling for quality conformance inspection shall be in accordance with MIL-STD-105, using the following Inspection Levels and Acceptable Quality Levels (AQL):

Inspection or Test	Inspection Level	AQL
Product examination (4.5)	II	1.0
Functional test (4.6.1)	II	1.0
Packaging inspection (4.7)	S-4	4.0

* 4.5 Examination. Visually and manually examine the timing light to determine conformance with the requirements of 3.2 through 3.3, 3.4 through 3.4.3 and 3.6 through 3.11. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the timing light has passed the

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examinations prescribed for them by the contractor's inspection system shall be cause for rejection.

4.6 Tests.

4.6.1 Functional test. With batteries installed, the timing light shall be turned on. Both glow lamps shall glow. Contact each red clip with the black clip for a period of one minute. Upon contact the corresponding lamp shall be extinguished. Turn the timing light off. After completion of the above, and while in an ambient air temperature of +70° to +90°F, one high tension lead of a magneto shall be connected to a three-point spark gap set at 0.010 inch and the magneto driven at a speed equivalent to a 4-cycle engine speed of 1,000 rpm. The timing light shall be properly connected to the magneto and turned on. The spark gap shall be observed and any evidence of sparking shall be cause for rejection.

4.6.2 Endurance tests.

4.6.2.1 Continuous operation test. The timing light, with batteries installed, shall be subjected to a minimum of 2.5 hours continuous operation with the leads connected to 0.3 ohm resistors. After completion of the test, the timing light shall be connected to a dual ignition system and used to time the magnetos to the engine and to synchronize the two magneto breakers. Failure of the timing light to perform these operations satisfactorily shall be cause for rejection (see 3.5.1 and 3.5.2.1).

4.6.2.2 Intermittent operation test. With the leads connected to a resistance of 0.3 ohms, the timing light, with batteries installed, shall be subjected to 380 cycles of one minute on and 15 seconds off. After completion of the test, the timing light shall be connected to a dual ignition system and used to time the magnetos with the engine and to synchronize the two magneto breakers. Failure of the timing light to perform these operations satisfactorily shall be cause for rejection (see 3.5.1 and 3.5.2.2).

4.6.3 Vibration test. The complete timing light, with batteries installed, shall be securely fastened to a vibration table and subjected to a vibration of simple harmonic motion having an amplitude of 0.030 inch (0.060 inch maximum total excursion), with the frequency being varied uniformly between the limits of five to 55 Hz. The entire frequency range from five to 55 Hz and back to five Hz shall be traversed in approximately one minute. Duration shall be for one hour in the direction of each of the three major axes. Following the test, the timing light shall be examined for evidence of damage and tested in accordance with 4.6.1 to determine compliance with 3.5.3.

4.6.4 Shock test. The timing light, with batteries installed and cover in place, shall be dropped three times, once through each of its three major axes, to a concrete floor from a height of eight inches. The timing light shall then be examined for evidence of damage and tested in accordance with 4.6.1 to determine compliance with 3.5.4.

4.6.5 Operating temperature tests. While the timing light is undergoing the operating tests of 4.6.5.1 and 4.6.5.2, the accessory leads (see 3.6.2.3) shall also be subjected to the same tests.

* 4.6.5.1 Low temperature operating test. The timing light, with batteries installed and leads attached, shall be exposed to an ambient air temperature of +30° ±5°F until the unit is stabilized. While at that stabilized temperature, the

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timing light shall be tested in accordance with 4.6.1 to determine compliance with 3.5.5.

* 4.6.5.2 High temperature operating test. The timing light, with batteries installed and leads attached, shall be exposed to an ambient air temperature of +110° ±5°F until the unit is stabilized. While at that stabilized temperature, the timing light shall be tested in accordance with 4.6.1 to determine compliance with 3.5.5.

* 4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

* 5.1 Preservation. Preservation shall be Level A or Industrial. Unless otherwise specified (see 6.2.1), Industrial preservation shall be provided. When military (Level A) preservation is required, it shall be so specified in the contract.

5.1.1 Level A.

5.1.1.1 Cleaning and drying. The timing light shall be cleaned and dried in accordance with the applicable procedures of MIL-P-116.

* 5.1.1.2 Unit packaging. The unit method of packaging shall conform to method IC-2 of MIL-P-116, using a fiberboard box conforming to PPP-B-636, class domestic, as the unit container. As an optional method, the carrying case may be substituted for the container required by method IC-2 provided that corners of the case and all projections are adequately cushioned to prevent damage to the barrier bag. Test leads shall be coiled to a minimum safe diameter and secured with cotton tape to prevent uncoiling. Place the leads in the storage compartment of the carrying case. Glow lamps shall be covered with 0.50 inch thick, uncompressed, cushioning material conforming to PPP-C-843, type II. Secure cushioning material with tape conforming to PPP-T-60, type and class optional. Place technical data, when applicable, in a plastic or kraft paper bag and position under the clip provided.

* 5.1.1.3 Intermediate packaging. The timing lights, packaged as specified in 5.1.1.2, shall be intermediate packaged in a close fitting fiberboard box conforming to PPP-B-636, grade WSC. The quantity of lights to be so packaged shall be governed by the weight limitations of the box specification. Closure of the box shall be as specified in the appendix to PPP-B-636.

* 5.1.2 Industrial. Industrial preservation shall be as specified in ASTM D3951-82.

* 5.2 Packing. Packing shall be Level A, Level B, or Industrial. Unless otherwise specified (see 6.2.1), Industrial packing shall be provided. When military (Levels A and B) packing is required, it shall be so specified in the contract.

* 5.2.1 Level A. Timing lights packaged as specified in 5.1.1.2 and 5.1.1.3 shall be packed in an exterior shipping container conforming to PPP-B-601, overseas type, or PPP-B-621, class 2. The gross weight of each shipping container shall not exceed the weight limitations of the applicable box specification. As

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far as practicable, containers shall be uniform in size and contain like quantities. Blocking, bracing, and cushioning shall be in accordance with MIL-STD-1186. Container closure and strapping requirements shall be in accordance with the appendix to the applicable box specification.

* 5.2.2 Level B. The timing lights shall be packed as specified in 5.2.1 except the shipping container shall conform to PPP-B-640, class 2.

* 5.2.3 Industrial. Industrial packing shall be as specified in ASTM D3951-82.

* 5.3 Marking. Marking shall be Level A, Level B, or Industrial. Unless otherwise specified (see 6.2.1), Industrial marking shall be provided. When military (Levels A and B) marking is required, it shall be so specified in the contract.

* 5.3.1 Level A and B marking. Level A and B marking shall be as specified in MIL-STD-129.

* 5.3.2 Industrial marking. Industrial marking shall be as specified in ASTM D3951-82.

6. NOTES

6.1 Intended use. The timing light covered by this specification is intended to be used for ignition timing and synchronization of magnetos. The timing light indicates the exact opening of contact points so that adjustments can be made by watching the instrument lights.

6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Exceptions and additional safety/health requirements when required (see 3.2.1 and 6.4).
(d) Optional arrangement of test leads when required (see 3.6.2.3).
(e) Painting and finishing if different (see 3.9).
(f) Product identification if different (see 3.10).
(g) First article inspection if different (see 4.3).
(h) Quality conformance inspection if different (see 4.4).
(i) Sampling if different (see 4.4.1).
(j) Level of preservation, packing, and marking if different (see 5.1, 5.2, and 5.3).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

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* 6.4 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:

Army - AL
Navy - YD
Air Force - 99

Preparing activity:

Army - AL

Project No. 4910-0539

Review activities:

Army - AV
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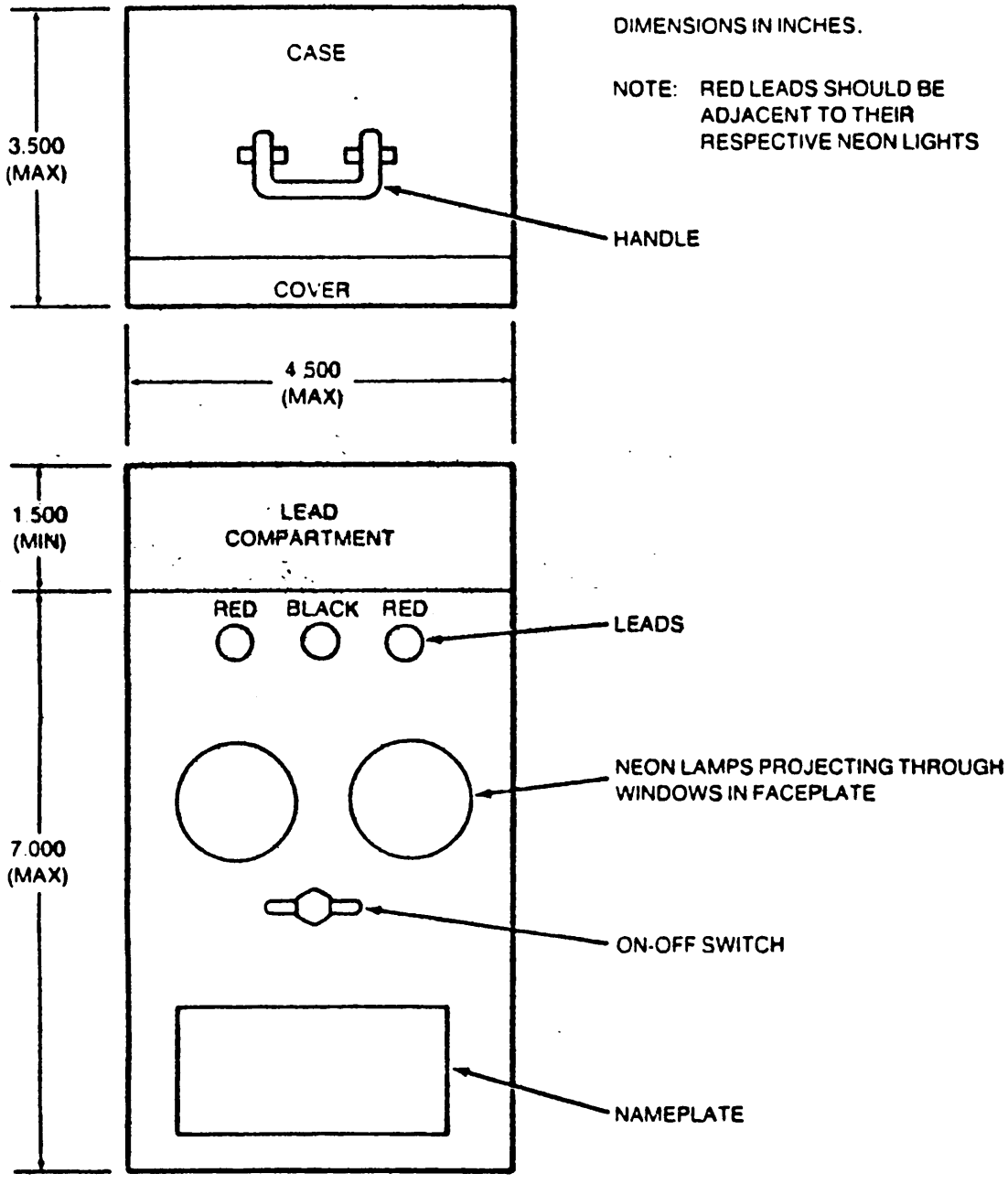


FIGURE 1. Case, magneto timing light.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-20703D	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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